Quantification: I534_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	26
interior lacuna count	23
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 23)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2954.39	2001	2699.77	px^2
major_axis_length_px	lacuna shape	84.9561	74.781	33.0167	px
minor_axis_length_px	lacuna shape	45.7378	37.2141	22.5561	px
aspect_ratio	lacuna shape	2.0152	1.9891	0.6188	unitless
eccentricity	lacuna shape	0.8185	0.8644	0.1167	unitless
solidity	lacuna shape	0.7412	0.724	0.0987	unitless
roots_count	headline	11.1739	10	5.8516	count
ring_length_r30_px	headline	295.217	277	143.733	px
ring_length_r60_px	reported	582.087	585	298.16	px
owned_length_px	ownership-dependent	715.849	506.031	630.617	px
edge_count	ownership-dependent	35.8261	23	30.28	count
mean_edge_length_px	ownership-dependent	19.3956	19.1762	4.6448	px
perimeter_px	normalised	281.136	270.635	124.881	px
ring_area_r30_px2	normalised	9267.43	8169	3530.85	px^2
ring_area_r60_px2	normalised	19156.3	17028	7217.71	px^2
in_frame_fraction_r30	normalised	0.9858	1	0.0341	unitless
in_frame_fraction_r60	normalised	0.9668	1	0.0743	unitless
ring_density_r30	normalised	0.0318051	0.0319518	0.0106386	px^-1
ring_density_r60	normalised	0.0300959	0.0316249	0.0104562	px^-1
roots_per_100px_perimeter	normalised	3.9781	4.137	1.1142	per 100 px
ring_attached_length_r30_px	network v2	272.217	232	144.292	px
ring_attached_length_r60_px	network v2	459.609	384	301.214	px
ring_length_w_r30_px	network v2	334.862	314.907	162.928	px
ring_length_w_r60_px	network v2	660.066	660.688	338.901	px
sholl_crossings_r10	network v2	6.7826	6	3.5156	count
sholl_crossings_r20	network v2	7.3478	6	4.2277	count
sholl_crossings_r30	network v2	8.1304	7	4.5458	count
loop_count	loops, report only	1.8261	0	2.9488	count
loop_edge_share	loops, report only	0.109	0	0.1485	share
ring_width_mean_r30_px	width	7.8422	7.7108	1.2862	px

Per field
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measure	value	units
analysed_area_px2	775235	px^2
total_skeleton_length_px	22893	px
canalicular_length_density_per_px	0.0295304	px^-1
junction_count	506	count
junction_density_per_px2	0.00065271	px^-2
lacuna_count	26	count

lacunae_per_px2	3.354e-05	px^-2
skeleton_component_count	273	count
mean_component_length_px	83.8571	px
median_component_length_px	22	px
field_length_density_w_per_px	0.033516	px^-1
field_density_without_flagged_per_px	0.0236897	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	148	px
unattached_component_count	10	count
canalicular_length_density_attached_per_px	0.0295304	px^-1
field_length_density_with_unattached_per_px	0.0297213	px^-1
excluded_area_px2	193353	px^2
analysed_field_area_px2	855223	px^2
loop_count	54	count
loop_edge_share	0.1401	share
edge_count	1385	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.1922	px
width_median_px	8.2462	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	23441	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	5018	113.612	66.0072	1.7212	0.8139	0.7134	-1.5671	33.3908	243.731	True
2	1429	75.9455	34.6321	2.1929	0.89	0.5788	0.9821	47.7124	162.336	False
3	1748	63.7004	52.4126	1.2154	0.5683	0.625	1.3536	36.7677	399.808	False
4	2001	74.781	36.9486	2.0239	0.8694	0.794	-1.1924	101.66	153.378	False
5	4592	90.8676	67.3599	1.349	0.6712	0.8423	-1.2953	137.617	765.059	False
6	902	70.1443	23.2471	3.0173	0.9435	0.6677	-1.2472	150.552	535.413	False
7	2333	67.9441	45.1639	1.5044	0.7471	0.9554	-0.4874	234.38	1000.91	True
8	11887	168.982	95.145	1.7761	0.8264	0.8606	-1.2985	286.803	681.158	False
9	697	50.8222	20.9969	2.4205	0.9107	0.7383	1.293	287.073	212.017	False
10	3532	87.4182	55.1855	1.5841	0.7756	0.7638	1.235	329.842	344.026	False
11	4129	107.816	50.0459	2.1543	0.8857	0.9077	-1.34	378.661	498.241	False
12	889	60.4502	27.413	2.2052	0.8913	0.6247	-1.2388	363.547	49.8526	False
13	1735	68.9793	37.2098	1.8538	0.842	0.6893	-0.903	463.884	594.474	False
14	751	42.1527	25.0435	1.6832	0.8044	0.8632	-1.1817	459.581	482.401	False
15	5417	114.374	70.6956	1.6178	0.7861	0.6168	1.316	527.301	55.7192	True
16	3037	99.839	58.2931	1.7127	0.8118	0.571	1.4007	558.019	301.395	False
17	4073	115.2	49.8465	2.3111	0.9015	0.7391	0.3504	603.966	867.145	False
18	2337	72.0848	52.1015	1.3835	0.6911	0.6974	-1.4357	607.677	651.435	False
19	1679	74.0209	37.2141	1.9891	0.8644	0.724	-0.9638	634.934	701.784	False
20	7591	130.597	95.8386	1.3627	0.6793	0.6789	-0.2363	683.477	82.1523	False
21	4140	142.999	43.9592	3.253	0.9516	0.7236	-1.1238	670.561	384.423	False
22	554	30.7716	24.9383	1.2339	0.5858	0.895	1.0673	729.574	697.603	False

23	640	57.661	17.8443	3.2313	0.9509	0.7018	-0.3969	793.93	358.258	False
24	2036	78.7104	35.4441	2.2207	0.8929	0.8879	-1.002	828.682	94.3777	False
25	1332	74.3708	26.5147	2.8049	0.9343	0.75	-1.2358	829.074	388.601	False
26	6240	115.676	84.3355	1.3716	0.6844	0.7226	-1.0938	881.946	274.251	False

Network per lacuna

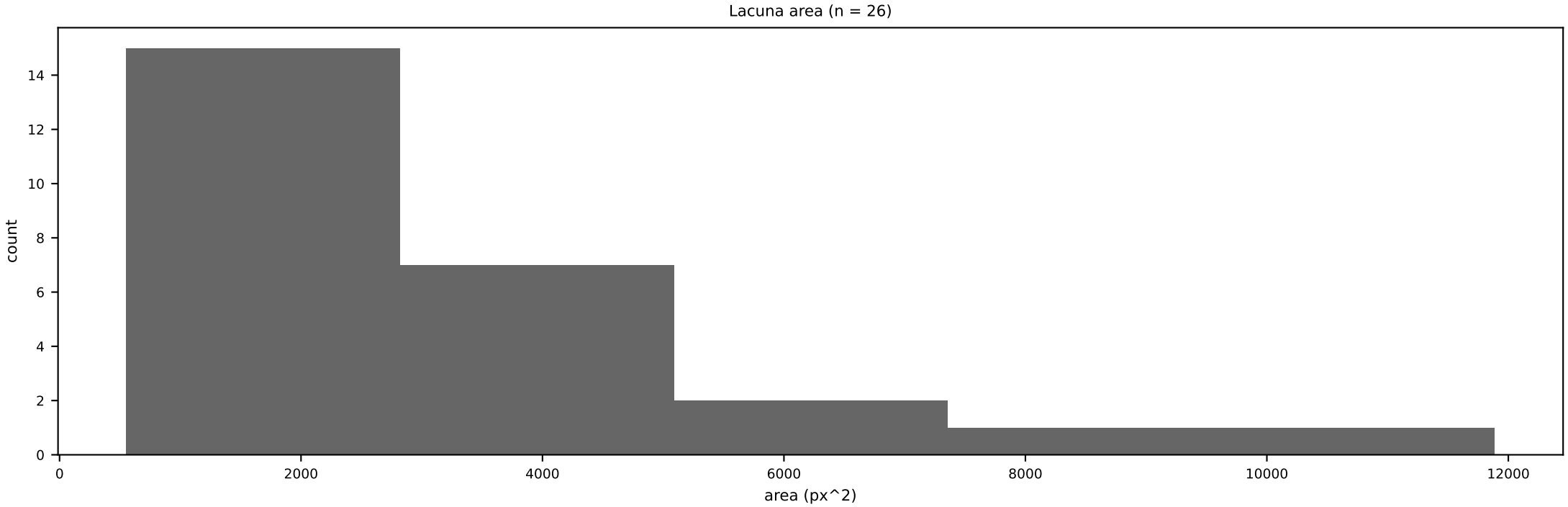
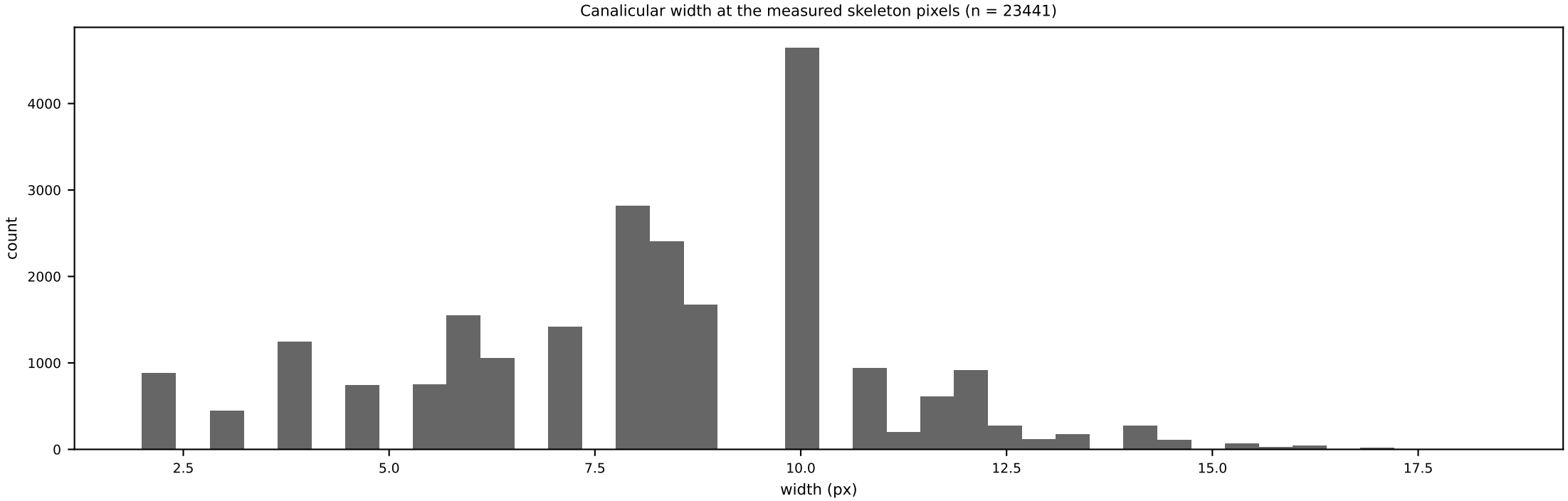
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	15	337	443	568.174	29	19.5922	6.7223
2	11	264	350	288.108	21	13.7194	6.4414
3	9	352	597	847.37	50	16.9474	7.3539
4	10	277	601	177.853	13	13.681	6.6277
5	15	438	737	1092.65	58	18.8388	7.5509
6	8	322	834	958.808	50	19.1762	7.7452
7	6	155	358	401.421	19	21.1274	8.7355
8	26	659	1235	2350.11	115	20.4357	8.1509
9	7	199	419	426.262	23	18.5331	8.5102
10	16	500	1055	2184.78	102	21.4194	8.5149
11	11	341	658	1054.25	55	19.1681	8.3777
12	8	192	330	232.836	10	23.2836	6.0936
13	12	360	668	1001.49	35	28.614	7.7108
14	5	155	258	175.125	10	17.5125	8.2917
15	19	351	508	723.027	34	21.2655	5.9354
16	19	385	862	1396.18	70	19.9455	7.8658
17	14	408	1045	1085.1	51	21.2766	7.1052
18	15	277	585	667.245	36	18.5346	7.6247
19	9	167	356	207.48	7	29.64	7.0953
20	23	499	869	903.436	46	19.6399	7.5211
21	9	207	349	267.61	17	15.7418	7.6147
22	5	175	403	152.882	7	21.8403	7.7313
23	1	35	62	7.8284	1	7.8284	12.9591
24	8	232	483	506.031	22	23.0014	8.1991
25	6	119	213	157.468	10	15.7468	7.2164
26	10	227	419	323.622	15	21.5748	8.0692

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	494.351	8231	12179	0.7421	0.6708	0.0409428	0.0363741	3.0343
2	282.291	4862	7142	0.9455	0.8134	0.0542986	0.0490059	3.8967
3	270.635	8169	16049	0.8695	0.7737	0.0430897	0.0371986	3.3255
4	236.936	6970	15047	1	1	0.0397417	0.0399415	4.2205
5	341.505	11247	24451	1	1	0.0389437	0.0301419	4.3923
6	176.167	7690	20930	1	1	0.0418726	0.0398471	4.5412
7	199.782	5730	14247	0.6804	0.6335	0.0270506	0.0251281	3.0033
8	548.257	17191	33534	1	1	0.038334	0.0368283	4.7423
9	136.125	6663	16736	1	1	0.0298664	0.0250358	5.1423
10	297.706	11030	23297	1	1	0.0453309	0.0452848	5.3744
11	305.605	10599	17028	1	1	0.0321729	0.0386422	3.5994
12	191.167	7524	16666	0.9189	0.7962	0.0255183	0.0198008	4.1848
13	238.451	9266	19320	1	1	0.0388517	0.0345756	5.0325
14	122.953	5564	11136	1	1	0.0278577	0.0231681	4.0666
15	490.399	11727	20563	0.8061	0.7376	0.0299309	0.0247046	3.8744
16	378.475	12733	27257	1	1	0.0302364	0.0316249	5.0201
17	373.546	12922	30830	1	1	0.0315741	0.0338956	3.7479
18	290.877	7526	16187	1	1	0.0368057	0.0361401	5.1568
19	233.38	6410	12295	1	1	0.026053	0.0289549	3.8564

20	555.955	15261	25089	0.9401	0.8614	0.0326977	0.0346367	4.137
21	383.889	12892	25468	1	1	0.0160565	0.0137035	2.3444
22	97.9828	5477	13592	1	1	0.0319518	0.0296498	5.1029
23	139.054	4710	7157	1	1	0.007431	0.00866285	0.7191
24	222.693	8786	20073	1	0.9917	0.0264056	0.0240622	3.5924
25	196.066	6012	12715	1	1	0.0197937	0.0167519	3.0602
26	446.416	13647	28597	1	1	0.0166337	0.0146519	2.2401

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	334	421	381.806	497.647	9	8	4
2	257	292	296.279	394.706	9	5	3
3	348	545	392.836	660.688	6	9	6
4	179	179	314.907	678.257	6	4	8
5	418	648	503.274	838.938	11	11	13
6	284	643	367.563	955.779	6	11	10
7	140	252	165.669	393.836	3	4	4
8	639	1187	726.487	1384.67	15	15	17
9	192	283	230.137	479.375	5	6	3
10	486	1025	568.688	1201.18	10	16	18
11	327	612	395.505	761.553	8	9	12
12	148	172	222.936	387.103	5	2	5
13	340	593	412.605	776.039	6	7	9
14	145	178	175.297	287.551	4	4	4
15	327	352	403.576	592.541	7	3	5
16	376	722	438.233	975.264	11	10	11
17	395	767	467.961	1172.69	8	11	12
18	232	384	316.179	665.186	3	6	7
19	153	183	181.853	401.919	6	3	3
20	451	748	578.399	1005.9	13	14	13
21	199	281	226.853	381.007	5	4	7
22	150	279	193.711	457.304	4	6	5
23	10	10	37.799	67.6985	0	1	2
24	232	429	267.794	549.902	6	6	7
25	119	159	135.397	236.238	2	2	3
26	181	252	251.137	462.576	7	7	9

I534_z11c1-2: distributions, pre-validation, px